CO1-AT1 ASSESSMENT 1

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COURESE CODE: DSA0502 - Query Processing for Data Science

35. Library Management

Aim

Analyse library book records by loading data from multiple formats, cleaning the data, merging related datasets, generating summary statistics, identifying the category with the highest number of books, and exporting the cleaned dataset.

Files Used

- Books.csv
- Authors.csv
- Books.json
- Books.xml

Steps

1. Load all datasets.
2. Inspect the datasets.
3. Remove duplicate records.
4. Handle missing values.
5. Validate data types.
6. Merge Books and Authors using **AuthorID**.
7. Calculate summary statistics.
8. Identify the category with the highest number of books.
9. Export the cleaned dataset.

Python Code:

```
import pandas as pd
# Load datasets
books = pd.read_csv("Books.csv")
authors = pd.read_csv("Authors.csv")
books_json = pd.read_json("Books.json")
books_xml = pd.read_xml("Books.xml")
# Inspect datasets
print("Books Dataset")
print(books.head())
print("\nAuthors Dataset")
print(authors.head())
# Remove duplicate records
books = books.drop_duplicates()
authors = authors.drop_duplicates()
# Handle missing values
books = books.dropna()
authors = authors.dropna()
# Validate data types
books["AuthorID"] = books["AuthorID"].astype(int)
authors["AuthorID"] = authors["AuthorID"].astype(int)
# Merge datasets using AuthorID
data = pd.merge(books, authors, on="AuthorID")
# Summary statistics
print("\nSummary Statistics")
print(data.describe(include="all"))
# Category-wise book count
category_count = data.groupby("Category")["BookID"].count()
# Category with highest number of books
highest_category = category_count.idxmax()
highest_value = category_count.max()
print("\nCategory-wise Book Count")
```

```
print(category_count)
print("\nCategory with Highest Number of Books:")
print(highest_category, "-", highest_value)
# Export cleaned dataset
data.to_csv("Cleaned_Library_Data.csv", index=False)
print("\nCleaned dataset exported successfully.")
```

OUTPUT:

```
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl) (2.0.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Books Dataset
   BookID                    Title  AuthorID     Category  Price
0       1            Python Basics       101  Programming    450
1       2  Data Science Essentials       102   Technology    600
2       3         Machine Learning       103   Technology    700
3       4         Database Systems       104     Database    550
4       5  Artificial Intelligence       103   Technology    800

Authors Dataset
   AuthorID     AuthorName    Country
0       101     John Smith        USA
1       102    Alice Brown         UK
2       103      David Lee     Canada
3       104    Emma Wilson  Australia
4       105  Michael Scott      India

Summary Statistics
          BookID          Title    AuthorID    Category       Price  \
count   6.000000              6    6.000000           6    6.000000
unique       NaN              6         NaN           3         NaN
top          NaN  Python Basics         NaN  Technology         NaN
freq         NaN              1         NaN           3         NaN
mean    3.500000            NaN  103.000000         NaN  600.000000
std     1.870829            NaN    1.414214         NaN  130.384048
min     1.000000            NaN  101.000000         NaN  450.000000
25%     2.250000            NaN  102.250000         NaN  512.500000
50%     3.500000            NaN  103.000000         NaN  575.000000
75%     4.750000            NaN  103.750000         NaN  675.000000
max     6.000000            NaN  105.000000         NaN  800.000000

       AuthorName Country
count           6       6
unique          5       5
top     David Lee  Canada
freq            2       2
mean          NaN     NaN
std           NaN     NaN
min           NaN     NaN
25%           NaN     NaN
50%           NaN     NaN
75%           NaN     NaN
max           NaN     NaN

Category-wise Book Count
Category
Database       1
Programming    2
Technology     3
Name: BookID, dtype: int64

Category with Highest Number of Books:
Technology - 3

Cleaned dataset exported successfully.
Saved at: Cleaned_Library_Data.csv
```

36. Online Orders

Aim

Analyze online order records by loading datasets from CSV, JSON, and XML formats, cleaning the data, merging related datasets, performing aggregation and summary statistics, identifying the category with the highest sales, and exporting the cleaned dataset.

Files Used

- Orders.csv
- Customers.csv
- Orders.json
- Orders.xml

Steps:

1. Load all datasets (CSV, JSON, and XML).
2. Inspect the datasets using head() and info().
3. Remove duplicate records.
4. Handle missing values by removing or filling them.
5. Validate data types of the columns.
6. Merge **Orders.csv** and **Customers.csv** using the common key **CustomerID**.
7. Perform aggregation and generate summary statistics.
8. Calculate the total sales for each product category.
9. Identify the category with the highest total sales.
10. Export the cleaned dataset to a new CSV file.

CODE:

```
# Install required library
!pip install lxml
import pandas as pd
# Load datasets
orders = pd.read_csv("Orders.csv")
customers = pd.read_csv("Customers.csv")
orders_json = pd.read_json("Orders.json")
orders_xml = pd.read_xml("Orders.xml")
# Inspect datasets
print("Orders Dataset")
```

```
print(orders.head())
print("\nCustomers Dataset")
print(customers.head())

# Remove duplicate records
orders = orders.drop_duplicates()
customers = customers.drop_duplicates()
# Handle missing values
orders = orders.dropna()
customers = customers.dropna()
# Validate data types
orders["CustomerID"] = orders["CustomerID"].astype(int)
customers["CustomerID"] = customers["CustomerID"].astype(int)
# Merge datasets
data = pd.merge(orders, customers, on="CustomerID")
# Summary statistics
print("\nSummary Statistics")
print(data.describe(include="all"))
# Category-wise Total Sales
category_sales = data.groupby("Category")["Amount"].sum()
# Highest value category
highest_category = category_sales.idxmax()
highest_value = category_sales.max()
print("\nCategory-wise Sales")
print(category_sales)
print("\nHighest Value Category")
print(highest_category, "-", highest_value)
# Export cleaned dataset
data.to_csv("Cleaned_Online_Orders.csv", index=False)
print("\nCleaned dataset exported successfully.")
```

OUTPUT:

```
Requirement already satisfied: lxml in /usr/local/lib/python3.12/dist-packages (6.1.1)
Orders Dataset
   OrderID  CustomerID     Category  Amount   OrderDate
0        1         201  Electronics   25000  2026-01-10
1        2         202      Fashion    3500  2026-01-12
2        3         203      Grocery    1800  2026-01-15
3        4         201  Electronics   15000  2026-01-20
4        5         204         Home    7000  2026-01-25

Customers Dataset
   CustomerID CustomerName       City
0         201        Rahul  Hyderabad
1         202        Priya    Chennai
2         203        Arjun  Bangalore
3         204       Sneha     Mumbai

Summary Statistics
         OrderID  CustomerID     Category        Amount   OrderDate  \
count   6.000000    6.000000            6      6.000000           6
unique       NaN         NaN            4           NaN           6
top          NaN         NaN  Electronics           NaN  2026-01-10
freq         NaN         NaN            2           NaN           1
mean    3.500000  202.166667          NaN   9416.666667         NaN
std     1.870829    1.169045          NaN   8945.259452         NaN
min     1.000000  201.000000          NaN   1800.000000         NaN
25%     2.250000  201.250000          NaN   3675.000000         NaN
50%     3.500000  202.000000          NaN   5600.000000         NaN
75%     4.750000  202.750000          NaN  13000.000000         NaN
max     6.000000  204.000000          NaN  25000.000000         NaN

       CustomerName       City
count             6          6
unique            4          4
top           Rahul  Hyderabad
freq              2          2
mean            NaN        NaN
std             NaN        NaN
min             NaN        NaN
25%             NaN        NaN
50%             NaN        NaN
75%             NaN        NaN
max             NaN        NaN

Category-wise Sales
Category
Electronics    40000
Fashion         7700
Grocery         1800
Home            7000
Name: Amount, dtype: int64

Highest Value Category
Electronics - 40000

Cleaned dataset exported successfully.
```